

Inspection Process Implementation Checklist

FRAPProject / ALBORAK

File-by-file implementation tracker

Status legend

- [x] Done
 - [] Pending
 - [~] Partially done / concept done, file not yet finalized in repo
 - N/A Not required yet
-

A. Models

A.1 Core inspection models

Master / setup

- ☒ Areas/AircraftMaintenance/Models/InspectionType.cs
- ☒ Areas/AircraftMaintenance/Models/MaintenanceProgram.cs
- ☒ Areas/AircraftMaintenance/Models/InspectionTypeProgram.cs
- ☒ Areas/AircraftMaintenance/Models/JobCard.cs
- ☒ Areas/AircraftMaintenance/Models/ProgramJobCard.cs
- ☒ Areas/AircraftMaintenance/Models/JobCardPlanningRule.cs
- ☒ Areas/AircraftMaintenance/Models/JobCardAttachment.cs

Runtime state

- ☒ Areas/AircraftMaintenance/Models/InspectionState.cs
- ☒ Areas/AircraftMaintenance/Models/AircraftJobCardState.cs

Workflow

- ☒ Areas/AircraftMaintenance/Models/WorkOrder.cs
- ☒ Areas/AircraftMaintenance/Models/WorkOrderJobCard.cs
- ☒ Areas/AircraftMaintenance/Models/WorkOrderJobCardSignOff.cs

A.2 Existing shared/base models reused

- ☒ Areas/Settings/Models/LookupBase.cs
 - ☒ Areas/Settings/Models/AcType.cs
 - ☒ Areas/Settings/Models/Aircraft.cs
 - ☒ Models/ApplicationUser.cs
-

A.3 Model notes

- ☒ InspectionType aligned to inherit from LookupBase
 - ☒ MaintenanceProgram aligned to inherit from LookupBase
 - ☒ Cleaned model architecture agreed
 - ☒ EntityConfiguration chosen as mapping source of truth
-

B. FRAContext

B.1 DbSet registrations

Inspection process DbSet

- ☒ DbSet<InspectionType> InspectionTypes
 - ☒ DbSet<MaintenanceProgram> MaintenancePrograms
 - ☒ DbSet<InspectionTypeProgram> InspectionTypePrograms
 - ☒ DbSet<JobCard> JobCards
 - ☒ DbSet<ProgramJobCard> ProgramJobCards
 - ☒ DbSet<JobCardPlanningRule> JobCardPlanningRules
 - ☒ DbSet<JobCardAttachment> JobCardAttachments
 - ☒ DbSet<InspectionState> InspectionStates
 - ☒ DbSet<AircraftJobCardState> AircraftJobCardStates
 - ☒ DbSet<WorkOrder> WorkOrders
 - ☒ DbSet<WorkOrderJobCard> WorkOrderJobCards
 - ☒ DbSet<WorkOrderJobCardSignOff> WorkOrderJobCardSignOffs
-

B.2 OnModelCreating registrations

Inspection process ApplyConfiguration(...)

- ☒ InspectionTypeConfiguration
 - ☒ MaintenanceProgramConfiguration
 - ☒ InspectionTypeProgramConfiguration
 - ☒ JobCardConfiguration
 - ☒ ProgramJobCardConfiguration
 - ☒ JobCardPlanningRuleConfiguration
 - ☒ JobCardAttachmentConfiguration
 - ☒ InspectionStateConfiguration
 - ☒ AircraftJobCardStateConfiguration
 - ☒ WorkOrderConfiguration
 - ☒ WorkOrderJobCardConfiguration
 - ☒ WorkOrderJobCardSignOffConfiguration
-

C. EntityConfiguration files

C.1 Completed configuration files

- ☒ Data/EntityConfigurations/InspectionTypeConfiguration.cs
- ☒ Data/EntityConfigurations/MaintenanceProgramConfiguration.cs
- ☒ Data/EntityConfigurations/InspectionTypeProgramConfiguration.cs
- ☒ Data/EntityConfigurations/JobCardConfiguration.cs
- ☒ Data/EntityConfigurations/ProgramJobCardConfiguration.cs
- ☒ Data/EntityConfigurations/JobCardPlanningRuleConfiguration.cs
- ☒ Data/EntityConfigurations/JobCardAttachmentConfiguration.cs
- ☒ Data/EntityConfigurations/InspectionStateConfiguration.cs
- ☒ Data/EntityConfigurations/AircraftJobCardStateConfiguration.cs

- ☒ Data/EntityConfigurations/WorkOrderConfiguration.cs
- ☒ Data/EntityConfigurations/WorkOrderJobCardConfiguration.cs
- ☒ Data/EntityConfigurations/WorkOrderJobCardSignOffConfiguration.cs

C.2 Configuration quality status

- ☒ Refactored to match cleaned models
 - ☒ Refactored to match LookupBase
 - ☒ SQL Server cascade-path issues resolved
 - ☒ Migration successfully applied
-

D. Migrations / Database

D.1 Migration files

- ☒ Migration created for inspection foundation
- ☒ Migration reviewed
- ☒ SQL Server constraint errors resolved
- ☒ Update-Database succeeded

Files

- ☒ Migrations/<timestamp>_AddInspectionProcessFoundation.cs
- ☒ Migrations/<timestamp>_AddInspectionProcessFoundation.Designer.cs

D.2 Database status

- ☒ New inspection foundation tables created
 - ☒ Keys created
 - ☒ FK constraints created
 - ☒ Unique indexes created
-

E. DTO / ViewModels

Status note: We designed these in detail conceptually. Depending on your repo, some

may still need to be physically created as files if not yet added.

E.1 Shared helper ViewModels

- [~] ViewModels/AircraftMaintenance/LookupOptionViewModel.cs
- [~] ViewModels/AircraftMaintenance/UserLookupViewModel.cs
- [~] ViewModels/AircraftMaintenance/AircraftLookupViewModel.cs
- [~] ViewModels/AircraftMaintenance/AcTypeLookupViewModel.cs

E.2 InspectionType ViewModels

- [~]
ViewModels/AircraftMaintenance/InspectionTypeListItemViewMode
l.cs
- [~]
ViewModels/AircraftMaintenance/InspectionTypeFormViewModel.cs
- [~]
ViewModels/AircraftMaintenance/InspectionTypeDetailsViewModel
.cs

E.3 MaintenanceProgram ViewModels

- [~]
ViewModels/AircraftMaintenance/MaintenanceProgramListItemView
Model.cs
- [~]
ViewModels/AircraftMaintenance/MaintenanceProgramFormViewMode
l.cs
- [~]
ViewModels/AircraftMaintenance/MaintenanceProgramDetailsViewM
odel.cs

E.4 JobCard ViewModels

- [~] ViewModels/AircraftMaintenance/JobCardListItemViewModel.cs
- [~] ViewModels/AircraftMaintenance/JobCardFormViewModel.cs
- [~] ViewModels/AircraftMaintenance/JobCardDetailsViewModel.cs
- [~]
ViewModels/AircraftMaintenance/JobCardAttachmentItemViewModel
.cs

E.5 JobCardPlanningRule ViewModels

- [~]
ViewModels/AircraftMaintenance/JobCardPlanningRuleListItemView
Model.cs
- [~]
ViewModels/AircraftMaintenance/JobCardPlanningRuleFormViewMod
el.cs
- [~]
ViewModels/AircraftMaintenance/JobCardPlanningRuleDetailsView
Model.cs

E.6 WorkOrder ViewModels

- [~]
ViewModels/AircraftMaintenance/WorkOrderListItemViewModel.cs
- [~] ViewModels/AircraftMaintenance/WorkOrderFormViewModel.cs
- [~] ViewModels/AircraftMaintenance/WorkOrderDetailsViewModel.cs
- [~]
ViewModels/AircraftMaintenance/WorkOrderJobCardItemViewModel.
cs
- [~]
ViewModels/AircraftMaintenance/WorkOrderJobCardSignOffItemVie
wModel.cs

E.7 Recommended grouping option

If grouped by feature instead of one file per class:

- ☐ ViewModels/AircraftMaintenance/SharedLookupViewModels.cs
- ☐ ViewModels/AircraftMaintenance/InspectionTypeViewModels.cs
- ☐
ViewModels/AircraftMaintenance/MaintenanceProgramViewModels.cs
- ☐ ViewModels/AircraftMaintenance/JobCardViewModels.cs
- ☐
ViewModels/AircraftMaintenance/JobCardPlanningRuleViewModels.
cs
- ☐ ViewModels/AircraftMaintenance/WorkOrderViewModels.cs

Choose either per-class files or grouped files, not both.

F. Repositories

F.1 Existing shared repository infrastructure

- ☒ Infrastructure/Interfaces/IGenericRepository.cs
- ☒ Infrastructure/Repositories/GenericRepository.cs

F.2 Existing UnitOfWork infrastructure

- ☒ Infrastructure/Interfaces/IUnitOfWork.cs
 - ☒ Infrastructure/UnitOfWork.cs or existing UnitOfWork implementation file
-

F.3 New inspection repositories — recommended next

InspectionType

- ☐
Areas/AircraftMaintenance/Repositories/IInspectionTypeRepository.cs
- ☐
Areas/AircraftMaintenance/Repositories/InspectionTypeRepository.cs

MaintenanceProgram

- ☐
Areas/AircraftMaintenance/Repositories/IMaintenanceProgramRepository.cs
- ☐
Areas/AircraftMaintenance/Repositories/MaintenanceProgramRepository.cs

JobCard

- ☐
Areas/AircraftMaintenance/Repositories/IJobCardRepository.cs
- ☐ Areas/AircraftMaintenance/Repositories/JobCardRepository.cs

JobCardPlanningRule

- ☐
Areas/AircraftMaintenance/Repositories/IJobCardPlanningRuleRepository.cs
- ☐

Areas/AircraftMaintenance/Repositories/JobCardPlanningRuleRepository.cs

WorkOrder

- ☐ Areas/AircraftMaintenance/Repositories/IWorkOrderRepository.cs
 - ☐ Areas/AircraftMaintenance/Repositories/WorkOrderRepository.cs
-

G. UnitOfWork updates

G.1 IUnitOfWork

Existing

- ☒ AcMainGroups repository property exists

Pending additions

- ☐ IInspectionTypeRepository InspectionTypes { get; }
- ☐ IMaintenanceProgramRepository MaintenancePrograms { get; }
- ☐ IJobCardRepository JobCards { get; }
- ☐ IJobCardPlanningRuleRepository JobCardPlanningRules { get; }
- ☐ IWorkOrderRepository WorkOrders { get; }

G.2 UnitOfWork implementation

Pending wiring

- ☐ instantiate InspectionTypeRepository
 - ☐ instantiate MaintenanceProgramRepository
 - ☐ instantiate JobCardRepository
 - ☐ instantiate JobCardPlanningRuleRepository
 - ☐ instantiate WorkOrderRepository
-

H. Controllers

H.1 Recommended first-wave controllers

InspectionType

- ☐ `Areas/AircraftMaintenance/Controllers/InspectionTypesController.cs`

MaintenanceProgram

- ☐ `Areas/AircraftMaintenance/Controllers/MaintenanceProgramsController.cs`

JobCard

- ☐ `Areas/AircraftMaintenance/Controllers/JobCardsController.cs`

JobCardPlanningRule

- ☐ `Areas/AircraftMaintenance/Controllers/JobCardPlanningRulesController.cs`

WorkOrder

- ☐ `Areas/AircraftMaintenance/Controllers/WorkOrdersController.cs`
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H.2 Suggested controller action checklist

InspectionTypesController

- ☐ `Index`
- ☐ `Details(int id)`
- ☐ `Create()`
- ☐ `Create(POST)`
- ☐ `Edit(int id)`
- ☐ `Edit(POST)`
- ☐ `Delete(int id)`
- ☐ `DeleteConfirmed(int id)`

MaintenanceProgramsController

- ☐ Index
- ☐ Details
- ☐ Create
- ☐ Edit
- ☐ Delete

JobCardsController

- ☐ Index
- ☐ Details
- ☐ Create
- ☐ Edit
- ☐ Delete

JobCardPlanningRulesController

- ☐ Index
- ☐ Details
- ☐ Create
- ☐ Edit
- ☐ Delete

WorkOrdersController

- ☐ Index
 - ☐ Details
 - ☐ Create
 - ☐ Edit
 - ☐ Open / Close / Status actions (later if needed)
 - ☐ Delete (optional depending on business rule)
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I. Views

I.1 InspectionType views

- ☐

Areas/AircraftMaintenance/Views/InspectionTypes/Index.cshtml

- ☐ Areas/AircraftMaintenance/Views/InspectionTypes/Create.cshtml
- ☐ Areas/AircraftMaintenance/Views/InspectionTypes/Edit.cshtml
- ☐ Areas/AircraftMaintenance/Views/InspectionTypes/Details.cshtml
- ☐ Areas/AircraftMaintenance/Views/InspectionTypes/Delete.cshtml

I.2 MaintenanceProgram views

- ☐ Areas/AircraftMaintenance/Views/MaintenancePrograms/Index.cshtml
- ☐ Areas/AircraftMaintenance/Views/MaintenancePrograms/Create.cshtml
- ☐ Areas/AircraftMaintenance/Views/MaintenancePrograms/Edit.cshtml
- ☐ Areas/AircraftMaintenance/Views/MaintenancePrograms/Details.cshtml
- ☐ Areas/AircraftMaintenance/Views/MaintenancePrograms/Delete.cshtml

I.3 JobCard views

- ☐ Areas/AircraftMaintenance/Views/JobCards/Index.cshtml
- ☐ Areas/AircraftMaintenance/Views/JobCards/Create.cshtml
- ☐ Areas/AircraftMaintenance/Views/JobCards/Edit.cshtml
- ☐ Areas/AircraftMaintenance/Views/JobCards/Details.cshtml
- ☐ Areas/AircraftMaintenance/Views/JobCards/Delete.cshtml

I.4 JobCardPlanningRule views

- ☐ Areas/AircraftMaintenance/Views/JobCardPlanningRules/Index.cshtml

- ☐ Areas/AircraftMaintenance/Views/JobCardPlanningRules/Create.cshtml
- ☐ Areas/AircraftMaintenance/Views/JobCardPlanningRules/Edit.cshtml
- ☐ Areas/AircraftMaintenance/Views/JobCardPlanningRules/Details.cshtml
- ☐ Areas/AircraftMaintenance/Views/JobCardPlanningRules/Delete.cshtml

I.5 WorkOrder views

- ☐ Areas/AircraftMaintenance/Views/WorkOrders/Index.cshtml
 - ☐ Areas/AircraftMaintenance/Views/WorkOrders/Create.cshtml
 - ☐ Areas/AircraftMaintenance/Views/WorkOrders/Edit.cshtml
 - ☐ Areas/AircraftMaintenance/Views/WorkOrders/Details.cshtml
 - ☐ Areas/AircraftMaintenance/Views/WorkOrders/Delete.cshtml
-

J. Mapping / helper layer

J.1 Entity ↔ ViewModel mapping

Can be manual in controllers first, or later extracted.

Pending

- ☐ InspectionType ↔ ViewModel mapping
- ☐ MaintenanceProgram ↔ ViewModel mapping
- ☐ JobCard ↔ ViewModel mapping
- ☐ JobCardPlanningRule ↔ ViewModel mapping
- ☐ WorkOrder ↔ ViewModel mapping

J.2 Optional later refactor

- ☐ dedicated mapper class(es)
- ☐ service layer for orchestration

K. Business logic services (later phase)

K.1 Not yet started

- ☐ inspection due calculation service
 - ☐ job-card planning rule evaluation service
 - ☐ work-order generation service
 - ☐ work-order closing service
 - ☐ signoff validation service
 - ☐ inspection state refresh service
 - ☐ aircraft job-card state recomputation service
-

L. Suggested implementation order from now

Phase 1 — first working CRUD slice

1. ☐ IInspectionTypeRepository
2. ☐ InspectionTypeRepository
3. ☐ update IUnitOfWork
4. ☐ update UnitOfWork
5. ☐ create InspectionTypesController
6. ☐ create InspectionTypes views
7. ☐ test full CRUD

Phase 2

1. ☐ MaintenanceProgram repository/controller/views
2. ☐ JobCard repository/controller/views

Phase 3

1. ☐ JobCardPlanningRule repository/controller/views

Phase 4

1. ☐ WorkOrder repository/controller/views
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M. High-level progress snapshot

Completed

- ☒ Models
- ☒ FRAContext DbSets
- ☒ EntityConfigurations
- ☒ Migration
- ☒ Database update

Designed but may still need file creation

- [~] DTO/ViewModels

Not yet implemented

- ☐ Repositories for inspection entities
 - ☐ UnitOfWork extensions
 - ☐ Controllers
 - ☐ Views
 - ☐ Mapping logic
 - ☐ Business services
-

N. Recommended immediate next file set

Do next

- ☐
Areas/AircraftMaintenance/Repositories/IInspectionTypeRepository.cs
 - ☐
Areas/AircraftMaintenance/Repositories/InspectionTypeRepository.cs
 - ☐ update Infrastructure/Interfaces/IUnitOfWork.cs
 - ☐ update Infrastructure/UnitOfWork.cs
 - ☐
Areas/AircraftMaintenance/Controllers/InspectionTypesController.cs
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